Comprehensive application

1. Course Content

Note: This program runs on a sandbox map; you will need to adjust the program for your personal map.

This tutorial uses the factory-installed road sign and lane recognition models, moving them center on a sandbox map. Upon recognizing a road sign, it performs the corresponding road sign function action.

2. Program Path

Autonomous Driving Source Code Path:
Lane Keeping:
`/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/auto_drive.py`

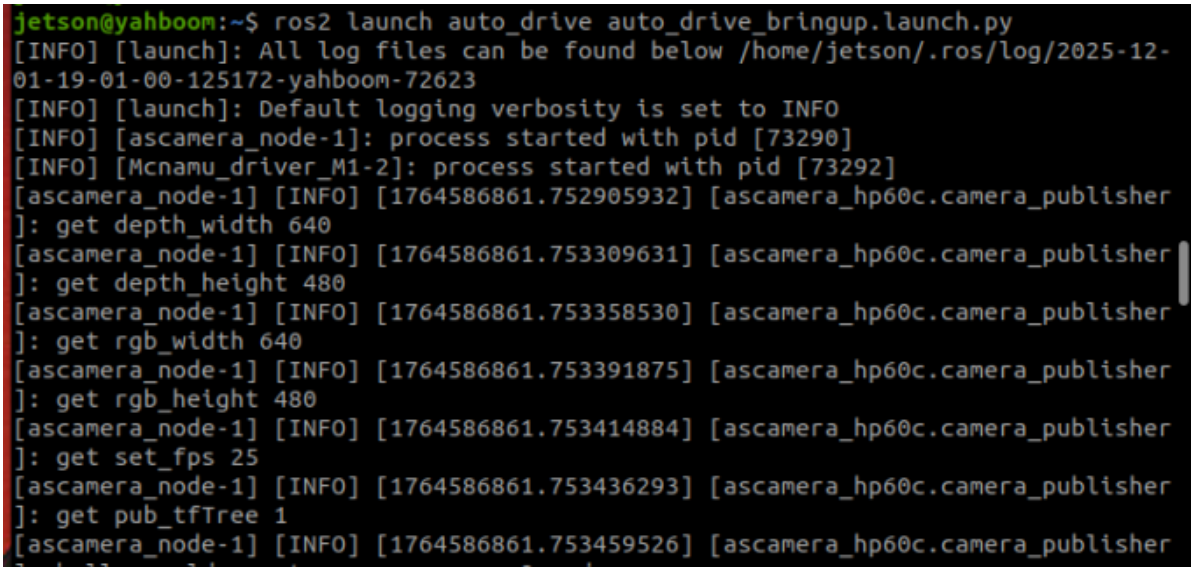
Road Sign Recognition:
`/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/yolo_detect.py`

3. Program Startup

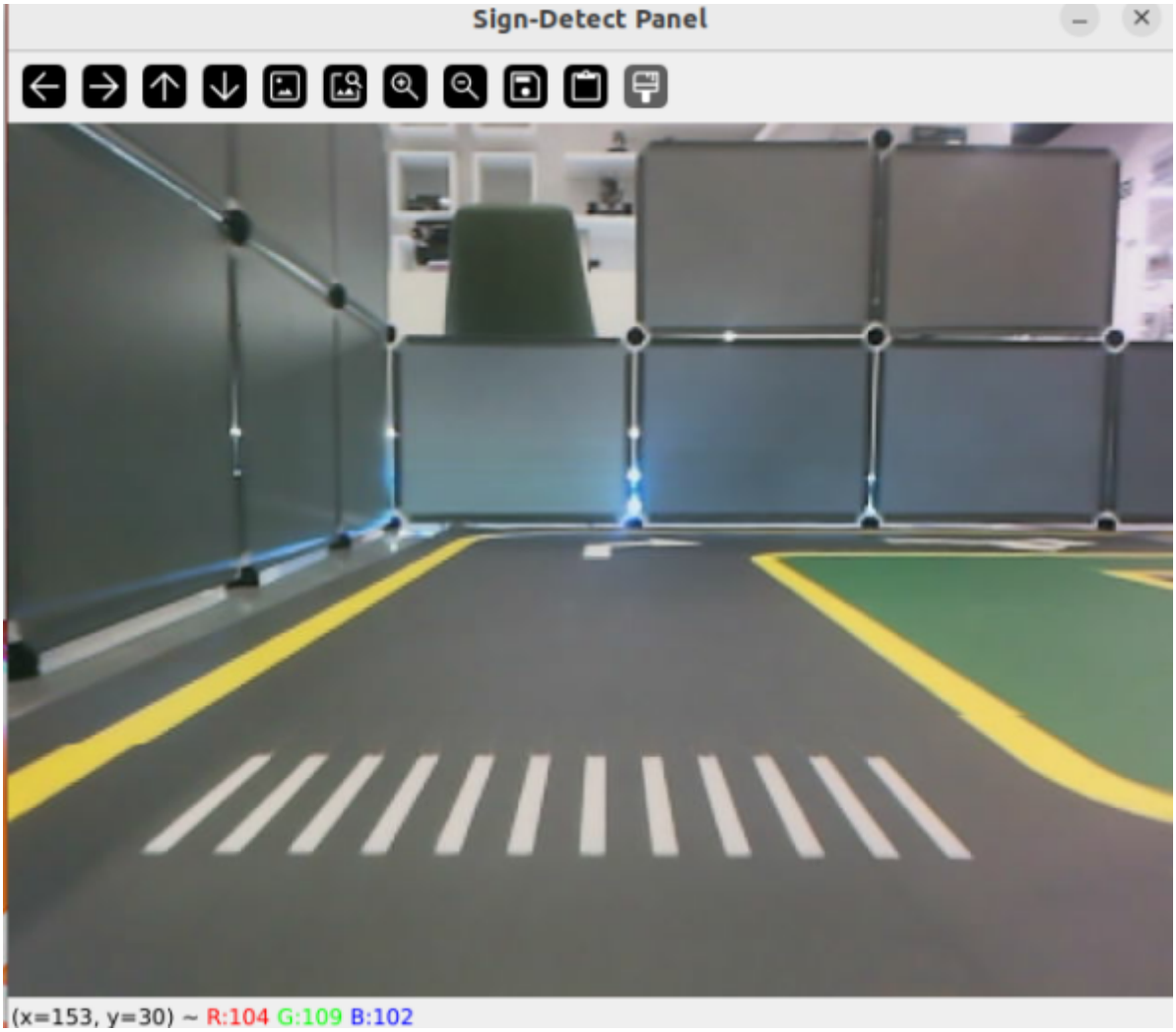
3.1 Startup Command

- Start camera + chassis + road signs and lane markings

```
ros2 launch auto_drive auto_drive_bringup.launch.py
```

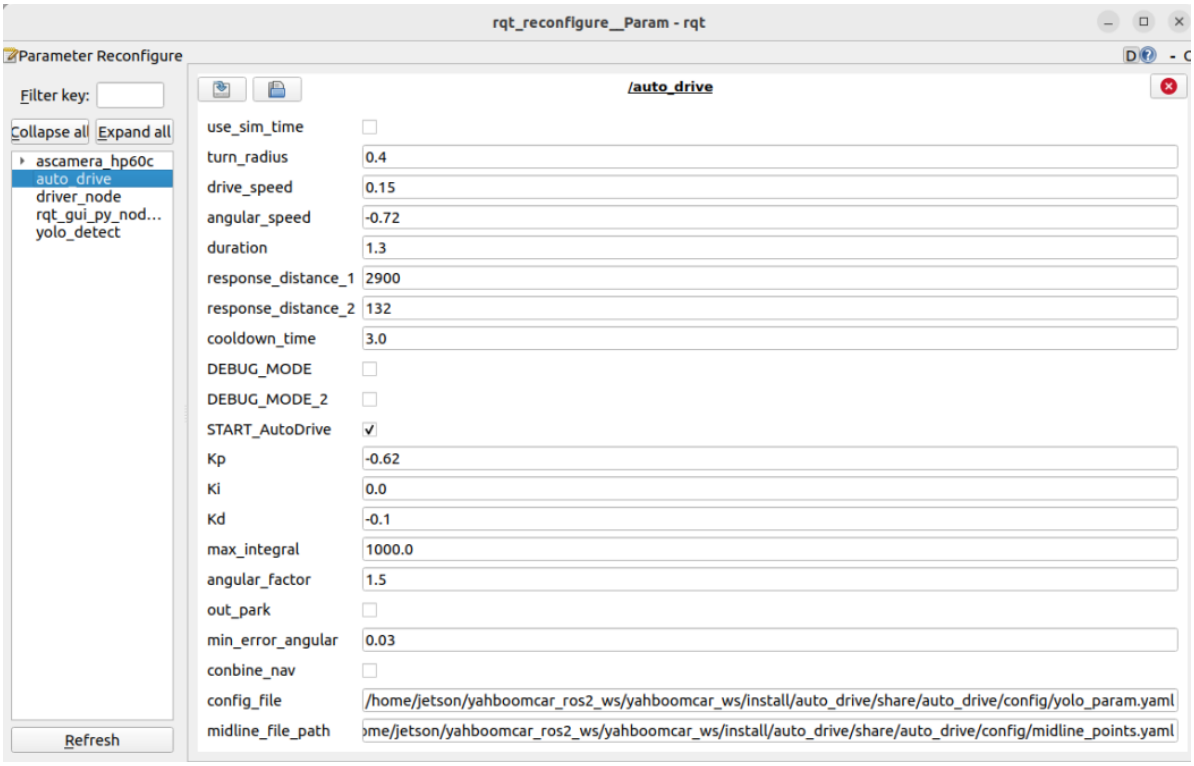


After the program runs, a YOLO recognition screen will be displayed. The car will move centered within the lane and perform corresponding actions based on the road signs it detects during its journey. (If the terminal does not report any errors after starting the program, but the car does not move, you can run the following program to check the parameters.)



- Start the dynamic parameter configuration node

```
ros2 run rqt_reconfigure rqt_reconfigure
```



If the car doesn't move, check two parameters: whether START_AutoDrive is checked and whether conbine_nav is false.

☑ After modifying the parameters, click in the blank area of the GUI to enter the parameter values. Note that this only applies to the current startup; for permanent effects, the parameters need to be modified in the source code.

🛠 Parameter Explanation:

【turn_radius】: Turning radius. Normally, this parameter does not need to be changed.

【drive_speed】: Linear velocity. The car's speed. The factory default is 0.15. If you modify this parameter, you need to adjust other parameters accordingly.

【angular_speed】: Turning angular velocity.

【duration】: Right turn time for recognizing right turn signs.

【cooldown_time】: Road sign recognition cooldown time.

【DEBUG_MODE】: Visual debug parameter. Enabling this may affect centering.

【START_AutoDrive】: Controls car movement parameters. True means the car can move.

【kp, ki, kd】: Car centering PID parameters.

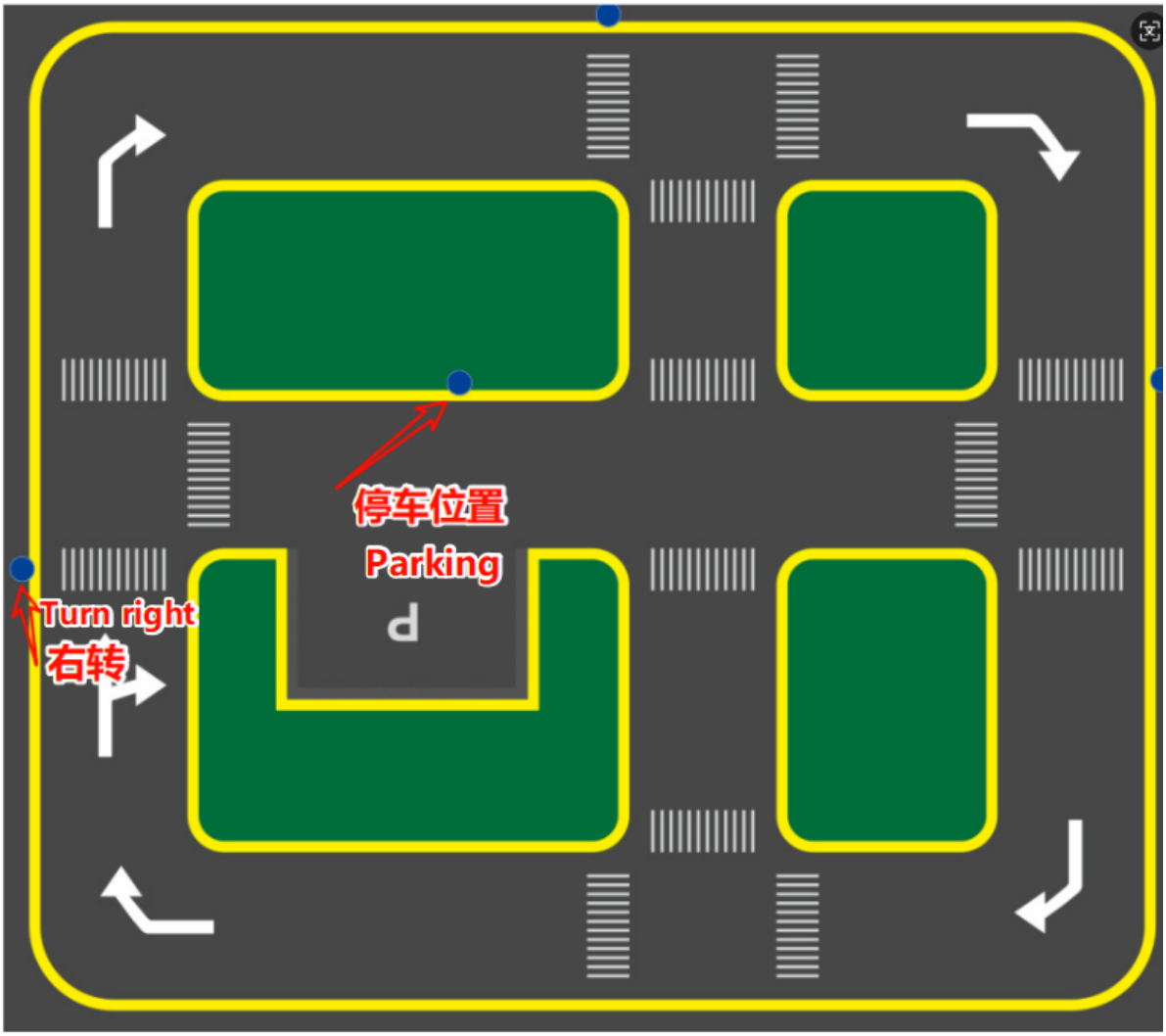
【turn_radius】: Turning radius. Normally, this parameter does not need to be changed.

【conbine_nav】: Required only when road network navigation is enabled and YOLO is activated. Enabling this will stop the car.

3.2 Car Movement After Recognizing Road Signs

- Straight ahead
  - Drive at the set speed, or stop and resume driving after encountering a parking sign
- Right turn
  - Trigger the fixed right turn action
- Horn
  - Trigger car horn sound effect
- Pedestrian crossing
  - Trigger car horn sound effect and reduce speed for 1 second
- Speed limit
  - Reduce car speed for 3 seconds, then return to normal speed
- School zone
  - Trigger car horn sound effect and reduce speed for 1 second
- Parking lots A and B
  - Trigger preset fixed parking action
- Traffic lights
  - Red light
    - Without road network planning, triggers disengagement of lane keeping assist.
    - With road network planning, cancels the current navigation task.
  - Green light
    - Without road network planning, triggers activation of lane keeping assist.
    - With road network planning, resumes the previous navigation task.

The right turn and parking positions are fixed. Since the program uses fixed actions, for better results, it is recommended to place them at these positions. If you need to place them elsewhere, you will need to readjust the fixed action timing.



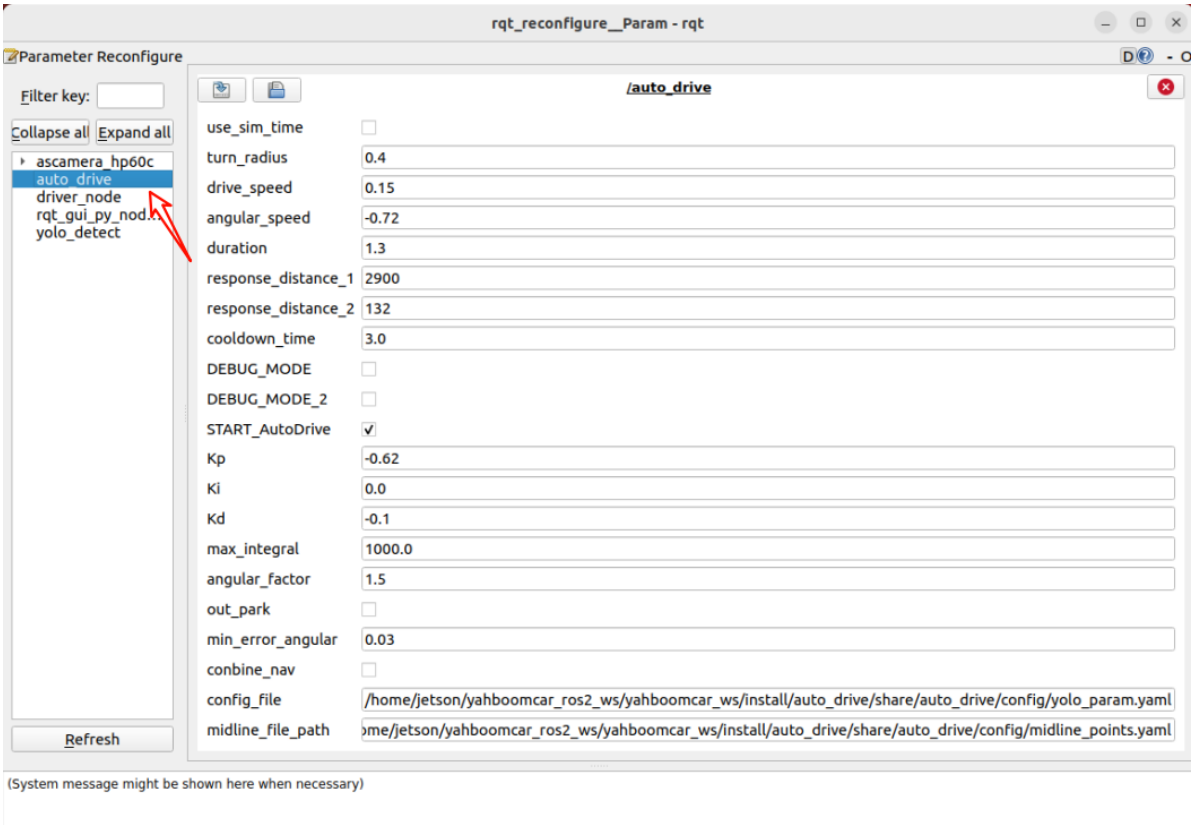
3.3 Debugging Parameters

There are two methods for debugging parameters: temporary real-time debugging and permanent parameter modification debugging.

- Temporary Real-Time Debugging

After running the program, run the dynamic parameter tuning command to debug:

```
ros2 run rqt_reconfigure rqt_reconfigure
```



Debugging Experience:

1. If lane centering is unstable, refer to the chapter "4. Lane Keeping in Autonomous Driving".
2. For right turn recognition, if the turn is insufficient or excessive, adjust the `duration` parameter to control the right turn time.
3. To prevent frequent road sign recognition, modify `cooldown_time`, which represents the cooldown time between road sign recognitions.
4. If lane centering adjustment is too slow, increase `kp`.

- Permanently Modify Parameters for Debugging

By modifying parameter files to achieve permanent parameter changes, the parameter paths are:

`/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/config/auto_drive_node.yaml`

`/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/config/yolo_param.yaml`

`/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/config/midline_points.yaml`

Three parameters represent: autonomous driving parameters, traffic sign recognition parameters, and lane centering parameters

Autonomous Driving Parameters:

```
auto_drive:
  ros__parameters:
    turn_radius: 0.4
    drive_speed: 0.15
    angular_speed: -0.72
    duration: 1.3
    response_distance_1: 2900
    response_distance_2: 132
    cooldown_time: 3.0
```

```
    DEBUG_MODE: False
    START_AutoDrive: True
    conbine_nav: False
    Kp: -0.62
    Ki: 0.0
    Kd: -0.1
    max_integral: 1000.0
    image_size: [640, 480]
```

The parameters above have the same meaning as the dynamically adjusted parameters. After modification, the program here will be called by default on the next run. Note the following during startup: `START_AutoDrive: True` and `conbine_nav: False`. These two parameters determine whether the car can run. `conbine_nav: False` is only required if road network navigation + autonomous driving is used.

Road Sign Recognition Parameters:

```
#Road Sign Detection Inference Parameters
sign_detection:
  model_path:
'/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/tools/sign_model.e
ngine'
  conf: 0.92     # Sets the minimum confidence threshold for detection. Detected
objects with confidence levels below this threshold will be ignored. Adjusting
this value helps reduce false positives.
  iou: 0.7      # The overlap threshold used for non-maximum suppression (NMS).
Lower values ••reduce detections by eliminating overlapping boxes.
  rect: True    # If enabled, performs minimum padding on the shorter side of
the image until it is divisible by the stride, improving inference speed.
  half: True    # If enabled, uses half-precision inference, which speeds up
model inference on supported GPUs with minimal impact on accuracy.
  device: "cuda:0"   # Inference device
  batch: 50          # Batch size. Larger batch sizes provide higher throughput,
thus reducing the total time required for inference.
  max_det: 4        # Maximum number of objects to detect
  augment: False    # Enables Test-Time Augmentation (TTA) for prediction, which
may improve detection robustness but slows down inference.
  verbose : False # Whether to output detailed inference information in the
terminal.
  classes : [0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13]

line_detection:
# Lane detection inference parameters
  model_path:
'/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/tools/lane_V6.engi
ne'
  conf: 0.5 # Sets the minimum confidence threshold for detection. Detected
objects with a confidence level below this threshold will be ignored. Adjusting
this value helps reduce false positives.
  iou: 0.6 # Overlap threshold for non-maximum suppression (NMS). Lower values
••reduce detections by eliminating overlapping boxes.
  rect: True # If enabled, performs minimum padding on the shorter side of the
image until it is divisible by the stride, improving inference speed.
  half: True # If enabled, uses half-precision inference, which speeds up model
inference on supported GPUs with minimal impact on accuracy.
  device: "cuda:0" # Inference device
```

```
  batch: 50 # Batch size. Larger batch sizes provide higher throughput, thus
reducing the total time required for inference.
  max_det: 4 # Maximum number of objects to detect
  augment: False # Enables test-time augmentation (TTA) for prediction, which
may improve detection robustness but slow down inference.
  verbose: False # Whether to output detailed inference information to the
terminal.
```

The above parameters can be used at the factory default settings and generally do not need to be modified.

Lane line centering parameters

```
midline_points:
  start_point:
    x: 360
    y: 477
  end_point:
    x: 355
    y: 424
  midline_length: 120
  mid_distance: [200,40]
```

The `start_point` and `end_point` parameters are used to calibrate the car's center point; see the course on centerline calibration for details. The `midline_length` parameter adjusts the car's turning timing on the sandbox map; a shorter value results in slower turns, and vice versa.

4. Source Code Analysis

4.1 auto_drive.py

Program source code path: `/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/auto_drive.py`

Control Flow consists of:

1. Receiving and processing image data
2. Detecting lane lines and traffic signs
3. Determining control strategy based on detection results
4. Issuing control commands (speed, direction)
5. Repeating the above steps

- Processing lane detection results

```
def __handle_lane_detect_result(self,left_box,right_box,frame):
        '''Processing lane detection results'''
        self.error_mode = 99
        error_angular = [0.0, 0.0]  # Default angle deviation

        if left_box is not None and right_box is not None and left_box[0]
<right_box[0] :
            # If both left and right lane lines exist, draw them and their
intersection
```

```
            self.error_mode=0  #Marking error modes: Centering mode: 0, Left-skewed mode: 1, Right-skewed mode: 2
            frame=self.lane_detect.draw_left_line(frame, left_box)# Draw left lane line
            frame=self.lane_detect.draw_right_line(frame, right_box)# Draw right lane line

 frame,x_center,y_center=self.lane_detect.draw_center_point(frame,left_box,right_box)# Draw estimated center point of the lane

 left_lane_offset,right_lane_offset=self.lane_detect.cal_lane_lateral_offset(left_box,right_box)
            # self.get_logger().info(f"left_lane_offset: {left_lane_offset:.2f}, right_lane_offset: {right_lane_offset:.2f} ")
            error_angular=self.lane_detect.cal_error_angular(x_center,y_center)# Calculate angular error between vehicle and center point

        elif left_box is not None and right_box is None and left_box[0]<320:
            self.error_mode=1
            self.lane_detect.prev_center=None
            frame=self.lane_detect.draw_left_line(frame, left_box)
        elif right_box is not None and left_box is None :
            self.error_mode=2
            self.lane_detect.prev_center=None
            frame=self.lane_detect.draw_right_line(frame, right_box)
        else:
            self.lane_detect.prev_center=None

        # When turn action is activated, temporarily suspend lane line speed control
        if self.turn_action_active:
            return frame

        if self.error_mode==0 :
            # Centering
            if error_angular[1]>=0.03 or error_angular[1]<=-0.03:
                if abs(error_angular[1])<0.3:
                    angular_z=self.pid_controller.pid_control(error_angular[1])
                    angular_z=min(angular_z,0.3)
                    angular_z=max(angular_z,-0.3)

 self.__set_current_speed(linear_x=self.drive_speed,angular_z=angular_z)


        elif self.error_mode==1:
            # Only left lane line exists
            if_infer=self.lane_detect.IF_lane_intersection(left_box)
            if if_infer is not None:

 self.__set_current_speed(linear_x=self.drive_speed,angular_z=self.angular_speed)
            else:

 self.__set_current_speed(linear_x=self.drive_speed,angular_z=0.0)
        else:
            self.__set_current_speed(linear_x=self.drive_speed,angular_z=0.0)

        # Display angular deviation on the image
```

```
        cv2.putText(frame, f"Angle Dev: {error_angular[1]:.2f} rad", (10, 90),
cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 0, 255), 2)
        frame=self.lane_detect.draw_midline(frame)
        return frame
```

- `auto_drive` node's `sign_callback` callback function subscribes to the `/sign_detect` topic, then executes actions corresponding to traffic signs in the handle_events thread.

```
    def handle_events(self):
        '''Handling target recognition events'''
        while True:
            try:
                object:Object=self.event_queue.get(timeout=0.2)
            except queue.Empty:
                continue
            if self.__check_cooldown(object.class_name):
                continue
            # self.get_logger().info(f'{object.class_name} handle')

            if object.class_name=="straight" and self.sign_enable_list[10]:
                self.go_straight()
            elif object.class_name=="turnright" and self.sign_enable_list[11]:
                self.turn('right')
            elif object.class_name=="turnright_ground":
                pass
            elif object.class_name=="honking" and self.sign_enable_list[1]:
                self.car_horn()
            elif object.class_name== "sidewalk" and self.sign_enable_list[6]:
                self.sidewalk()
            elif object.class_name=="sidewalk_ground" and
self.sign_enable_list[7]:
                pass
            elif object.class_name=="speedlimit" and self.sign_enable_list[8]:
                self.speedlimit()
            elif object.class_name=="stop" and self.sign_enable_list[9]:
                self.stop()
            elif object.class_name=="school" and self.sign_enable_list[5]:
                self.school()
            elif object.class_name=="parkA" and self.sign_enable_list[2]:
                    self.parking_A()
            elif object.class_name=="parkB" and self.sign_enable_list[3]:
                self.parking_A()
            elif object.class_name=="greenlight" and self.sign_enable_list[0]:
                self.greenlight()
            elif object.class_name =="redlight" and self.sign_enable_list[4]:
                self.redlight()
            elif object.class_name=="yellowlight" and self.sign_enable_list[13]:
                pass
            elif object.class_name=="out_park" :
                self.out_parking()
```

4.2、yolo_detect.py

Program source code is located at `/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/yolo_detect.py`

Control Flow

1. **Image Callback** (image_callback):
    - Convert ROS image message to OpenCV format
    - Put image into processing queue
2. **Sign Detection Processing** (handle_sign):
    - Get image frame from queue
    - Perform object detection using YOLO model
    - Process detection results:
        - Filter out ignored signs (e.g., "turnright_ground", "sidewalk_ground")
        - Publish detected sign information
        - Display FPS and inference time in debug mode
        - Show detection results in real-time

```
def handle_sign(self):
        '''Process road sign detection results in the image'''
        while True:
            try:
                frame = self.sign_detect.image_queue.get(timeout=0.01)
            except queue.Empty:
                continue
            infer_start = time.time()
            results=self.sign_detect.get_infer_result(frame,True)
            for res in results:
                if res.boxes is not None:
                    annotated_frame = res.plot()
                    boxes = res.boxes
                    for i in range(len(boxes)):
                        box_coords = boxes.xyxy[i].tolist()# Get bounding box
coordinates (xyxy format: xmin, ymin, xmax, ymax)
                        class_name = res.names[int(boxes.cls[i].item())]# Get
object name
                        confidence = boxes.conf[i].item()# Get confidence score

                        # Check if it's an ignored sign
                        if class_name in self.ignored_signs:
                            # rospy.loginfo(f"Ignoring sign: {class_name}")
                            continue  # Skip ignored signs


                        # Publish detection results
                        if self.sign_pub_counter % 5 == 0:
                            self.sign_publisher.publish(DetectionObject(
                                class_name=class_name,
                                confidence=confidence,
                                xmin=float(box_coords[0]),
                                ymin=float(box_coords[1]),
                                xmax=float(box_coords[2]),
```

```
                                ymax=float(box_coords[3])
                            ))
                        self.sign_pub_counter += 1

            if self.DEBUG_MODE:
                infer_end = time.time()
                # Update FPS
                self.frame_count += 1
                curr_time = time.time()
                time_diff = curr_time - self.prev_time
                if time_diff >= 1.0:
                    self.fps = self.frame_count / time_diff
                    self.frame_count = 0
                    self.prev_time = curr_time
                # Display FPS on the image
                cv2.putText(annotated_frame,f"FPS: {self.fps:.2f}",(10,
25),cv2.FONT_HERSHEY_SIMPLEX,0.6,(0, 255, 0),1,)
                # Display inference time
                infer_time = (infer_end - infer_start) * 1000
                cv2.putText(annotated_frame, f"Infer: {infer_time:.1f} ms", (10,
45),cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 255), 1)

            cv2.imshow("Sign-Detect Panel", annotated_frame)
            cv2.waitKey(1)
```

The main thread is responsible for receiving images, while worker threads are responsible for processing images and performing detection tasks. Inter-thread communication is achieved through a queue.

4.2、yolov11_infer.py

The program's source code is located at `/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/utils/yolov11_infer.py`

Control Flow

Sign_Detect class:

- When creating a Sign_Detect instance, load and parse the configuration file.
- Call the init_yolo_engine() method:
  - Initialize the image queue (maximum capacity 10)
  - Set model inference parameters (confidence, IOU threshold, etc.)
  - Load the YOLO model
- Add the image to the processing queue using the put_image method
- Obtain the inference result using the get_infer_result method
- Perform object detection using the YOLO model

Lane_Detect class:

- **Lane line detection**: Detects lane lines in the image using the YOLO model.
- **Lane center point calculation**: Calculates the center points of the left and right lane lines.
- **Lane departure detection**: Calculates the vehicle's offset angle and distance relative to the lane.
- **Visualization**: Provides various plotting methods for displaying detection results.

```
class Sign_Detect(YOLO_MODEL):
    """Road sign detection"""
    def __init__(self,config_file_path):
        with open(config_file_path, "r") as file:
            full_config = yaml.safe_load(file)
            self.config_param = full_config.get("sign_detection", {})
...

class Lane_Detect(YOLO_MODEL):
    """Lane detection model"""
    def __init__(self,config_file_path:str,
                 midline_file_path:str):
        with open(config_file_path, "r") as file:
            full_config = yaml.safe_load(file)
            self.config_param = full_config.get("line_detection", {})
        with open(midline_file_path, "r") as file:
            self.point_config = yaml.safe_load(file)
...
```